

A Comprehensive Analysis of Gamified vs Laboratory Tasks for Selective Attention Assessment

BRSM PROJECT POSTER
PRESENTATION SP 2026

ABSTRACT

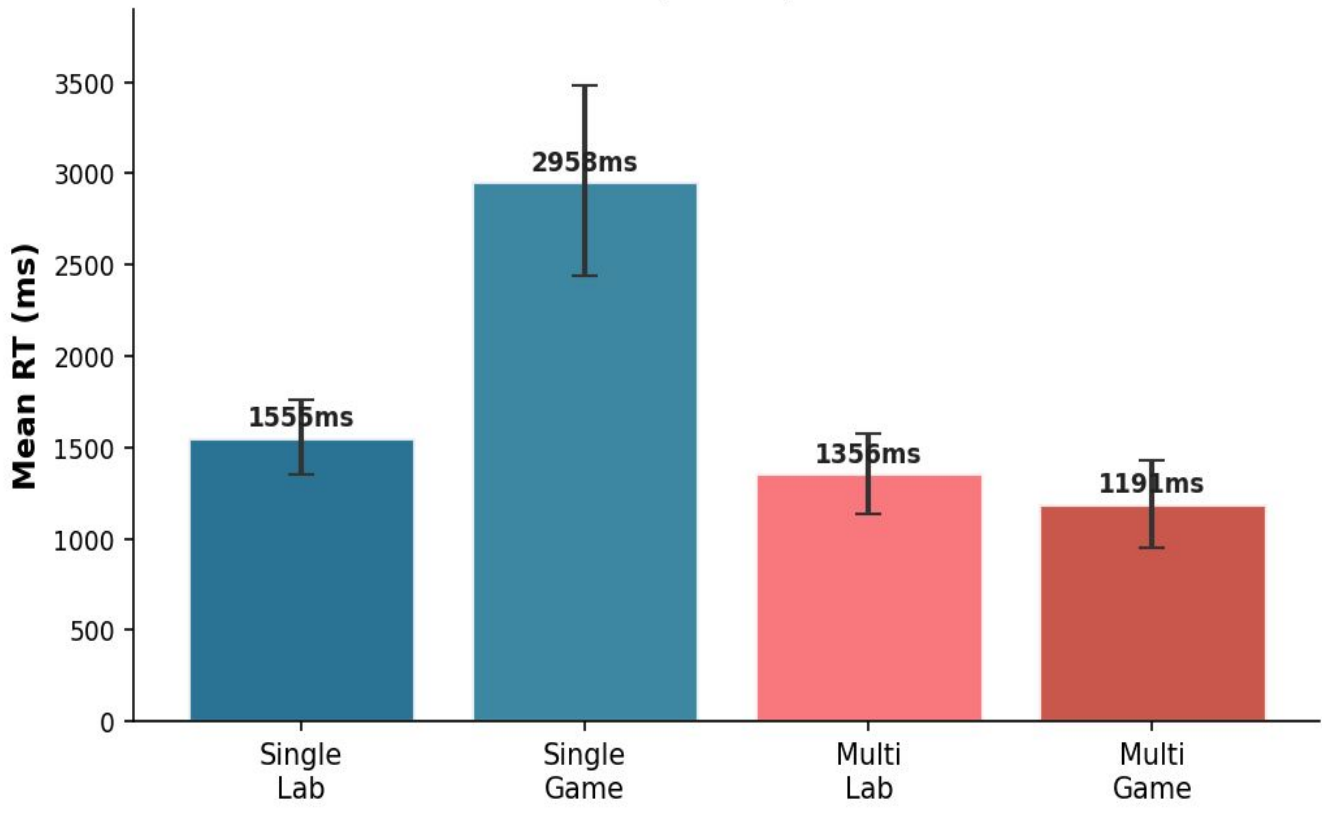
This study validates a smartphone-based gamified selective attention task against a standard desktop laboratory task by successfully differentiating cognitive load conditions ($p<.001$). A significant Modality $\times$ Load interaction ($F=140.8$, $p<.001$) revealed that the gamified version amplified overall task difficulty compared to the lab setting. Level-progression analysis highlighted escalating task difficulty and fatigue in the Single-Target game, evidenced by significant reaction time increases ($p<.001$). Conversely, the Multi-Target game demonstrated stable performance over time, characterized only by a slight, non-significant learning effect. These distinct performance findings demonstrate the gamified task's strong promise as a valid, accessible, and effective cognitive screening tool.

METHODOLOGY

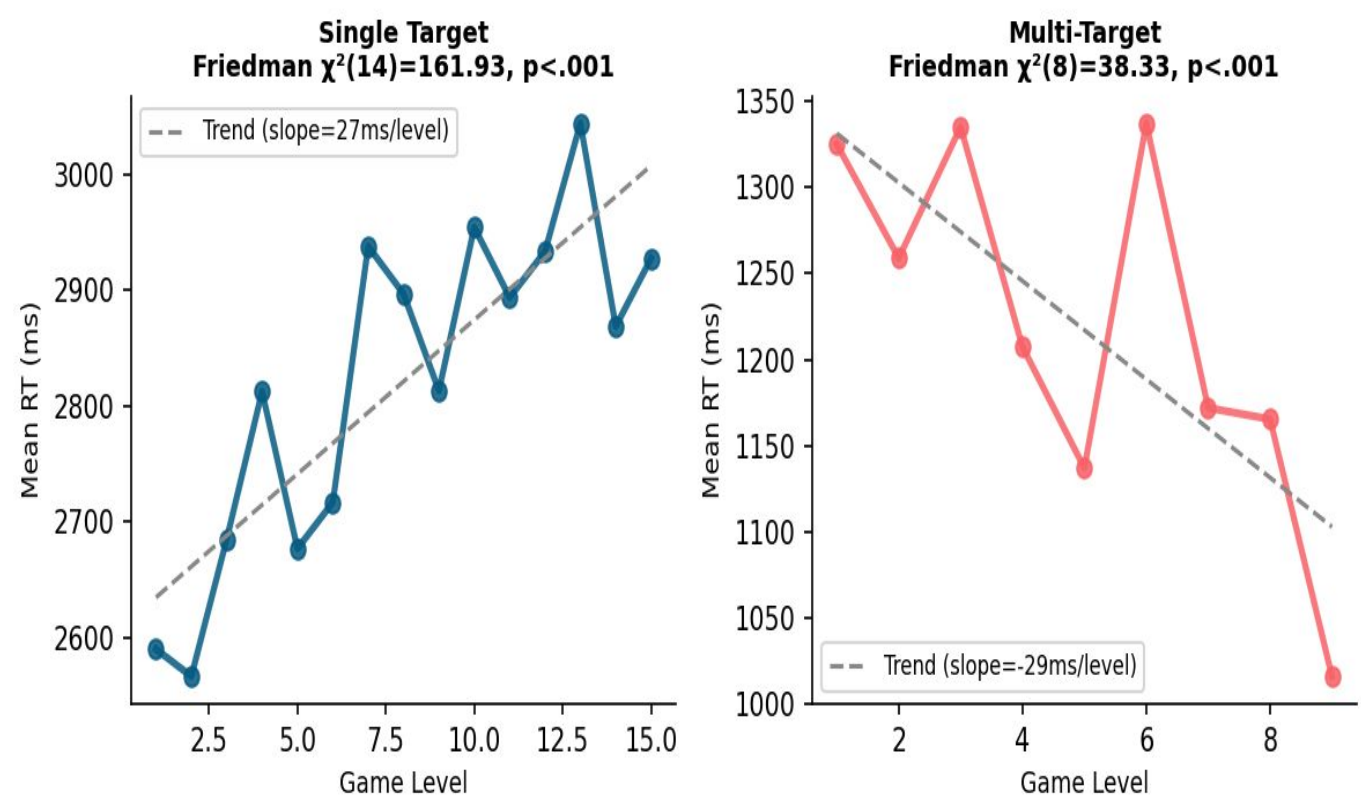
We employed a 2x2 Mixed Factorial design involving 37 participants. The between-subjects factor was Target Load, dividing participants into Single-Target ($n=21$) and Multi-Target ($n=16$) groups. The within-subjects factor was Modality, meaning all participants completed both a desktop-based Laboratory task and a smartphone-based Gamified task. We logged Reaction Time (RT) and Accuracy, analyzing the data using a 2x2 Mixed Factorial ANOVA, OLS regression modeling, and non-parametric hypothesis tests (Mann-Whitney, Wilcoxon, Friedman) to evaluate concurrent validity and task complexity.

RESULTS

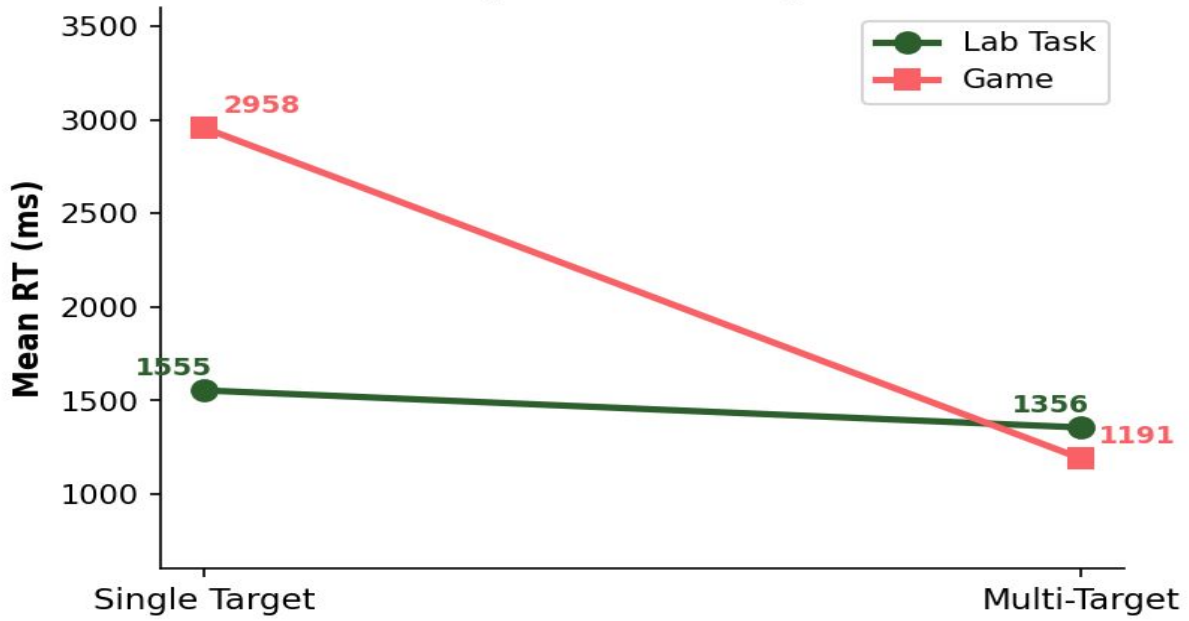
Reaction Time by Condition (± 1 SD)



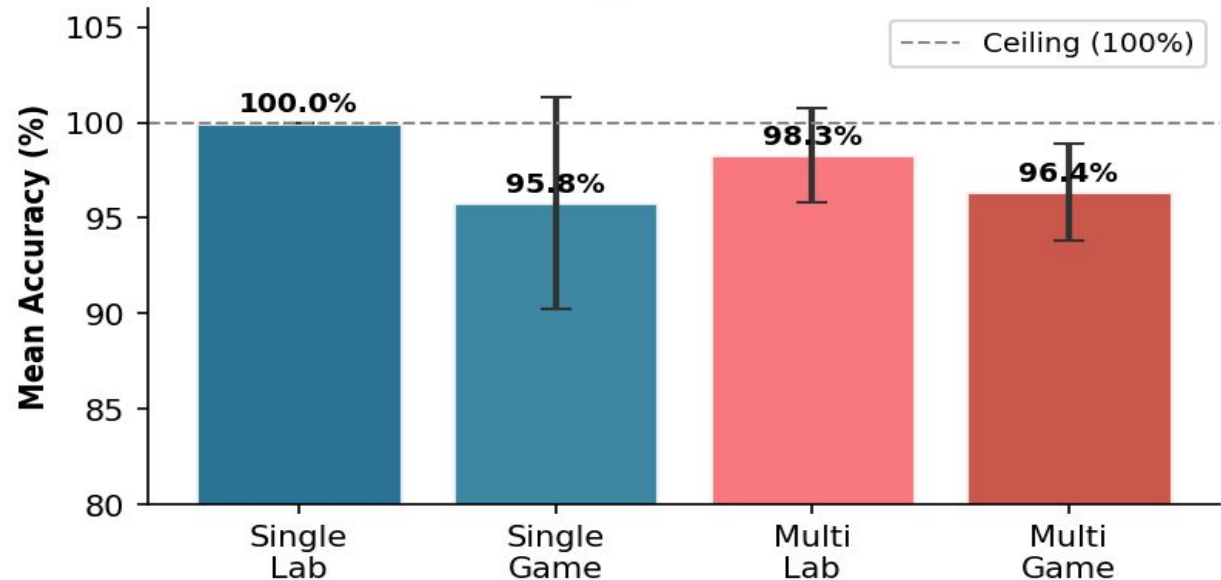
RQ4: Effect of Game Level on Reaction Time



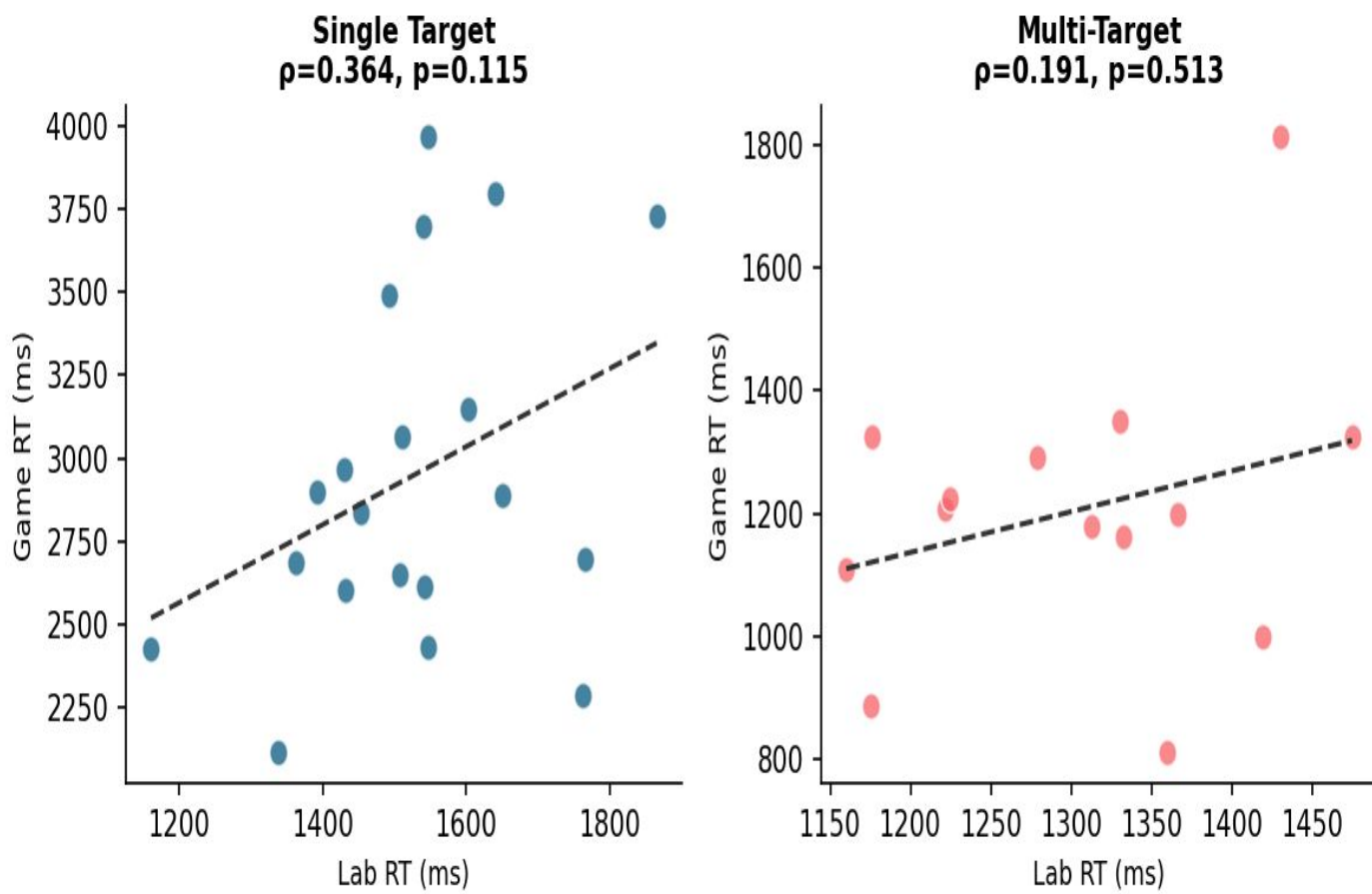
Modality $\times$ Load Interaction (Mixed ANOVA)



Accuracy by Condition (Ceiling Effect in Lab)



RQ1: Concurrent Validity – Game vs. Lab RT



CONTRIBUTIONS

- Pranav(2025201021): Exploratory Data Analysis (EDA), Research Question 4 (RQ4), OLS (RT_LOG Model), OLS (Level Regression)
- Vaibhav(2025201067): Research Questions 2 and 3 (RQ2, RQ3), ANOVA Analysis
- Afzal(2025201097): Project Design and Planning, Research Question 1 (RQ1), Paired t-test, Poster and Report Preparation

REFERENCES

- Waskom, M. L. (2021).seaborn: Statistical data visualization. Journal of Open Source Software, 6(60), 3021.
<https://doi.org/10.21105/joss.03021>
- Vallat, R. (2018). Pingouin: Statistics in Python.Journal of Open Source Software, 3(31), 1026.
<https://doi.org/10.21105/joss.01026>

Statistical Results Summary: RQ Analysis and Modeling

#	Research Question	Test / Method	Key Findings & Metrics	Decision
Hypothesis tests				
RQ1	Correlations	Spearman	Single: $p = .396$, $p = .075$ Multi combined: $p = .035$, $p = .923$ (ns) Overall combined: $p = .621$	Partial support
RQ2	Group Differences	Mann–Whitney	Lab: $U = 286$, $p < .001$ Game: $U = 336$, $p < .001$ Single > Multi	Reject H_0
RQ3	Within-Subject Change (Wilcoxon Ranked Signed-Rank)	Wilcoxon Ranked Signed-Rank	Single: $p < .001$ (Game > Lab) Multi: $p = .039$ (small diff)	Reject H_0
RQ4	Effect of Level (Friedman)	Friedman	Single: $\chi^2 = 161.93$, $p < .001$ Multi: $\chi^2 = 38.33$, $p < .001$	Reject H_0
Models				
5	ANOVA Mixed Design (Modality $\times$ Load)	2x2 Mixed Factorial ANOVA	Load: $F = 166.6$, $p < .001$ Modality: $F = 124.5$, $p < .001$ Interaction: $F = 140.8$, $p < .001$	Strong interaction
6	OLS (Reaction Time)	OLS Regression (RT_log DV)	$R^2 = .866$, $p < .001$ Excellent model fit	Significant model
7	OLS (Complexity Level)	OLS Regression (Game Level DV)	$R^2 = .385$, $p < .001$ $\uparrow$ RT (Single) $\downarrow$ RT (Multi)	Significant predictors
Learning Effect (MP 5 vs 5)				
8	Learning Effect (MP 5 vs 5)	Paired Sample t-test (Lab Modality)	$p = 0.586$ (ns) $\downarrow$ RT (Learning) Shapiro-Wilk: $p = 0.335$	Learning effect
9	Complexity (SP 8 vs 7)	Paired Sample t-test (SP modality)	$p < 0.001$ (***) $\uparrow$ RT (Task Diff.) Shapiro-Wilk: $p = 0.798$	Strong effect